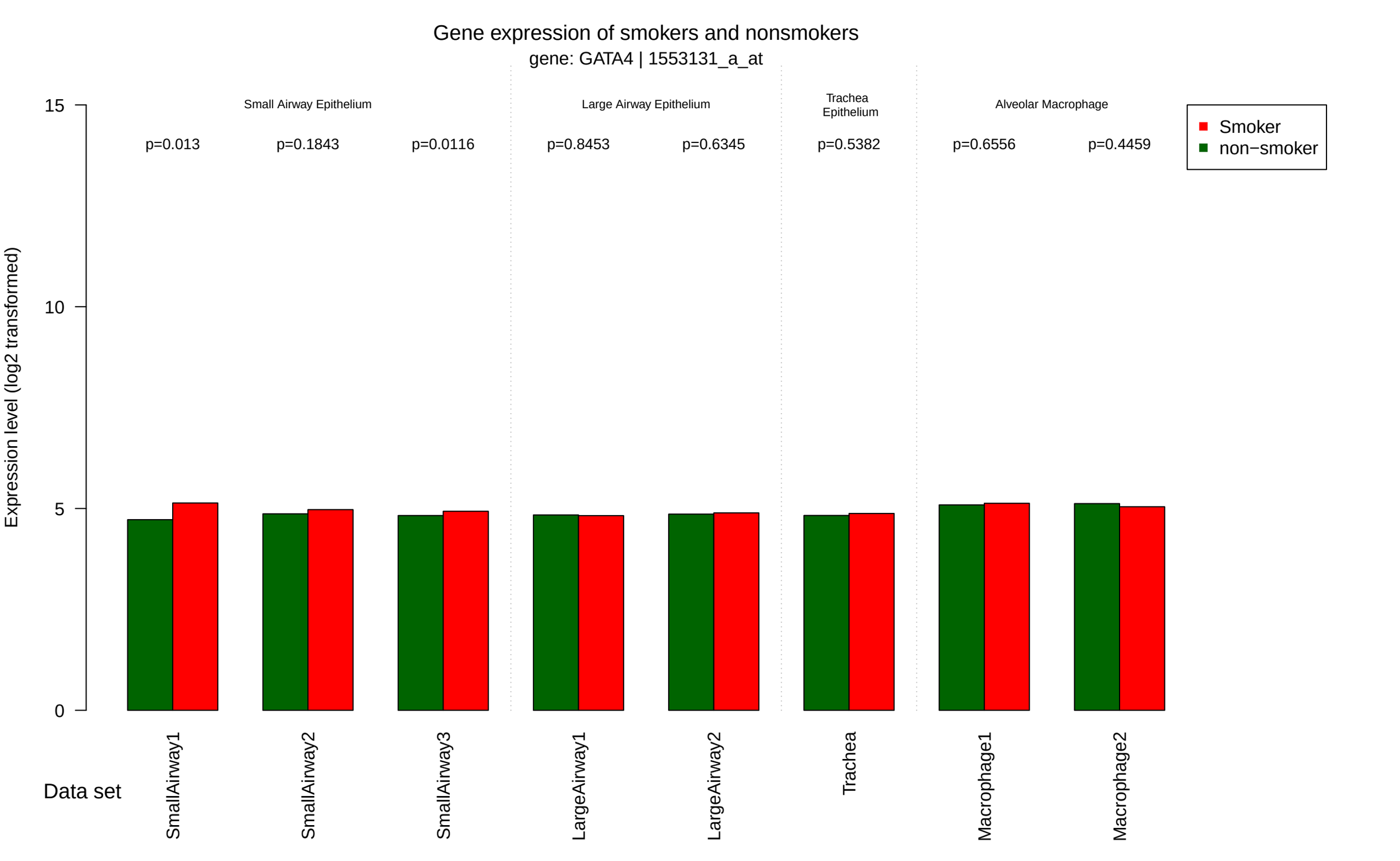
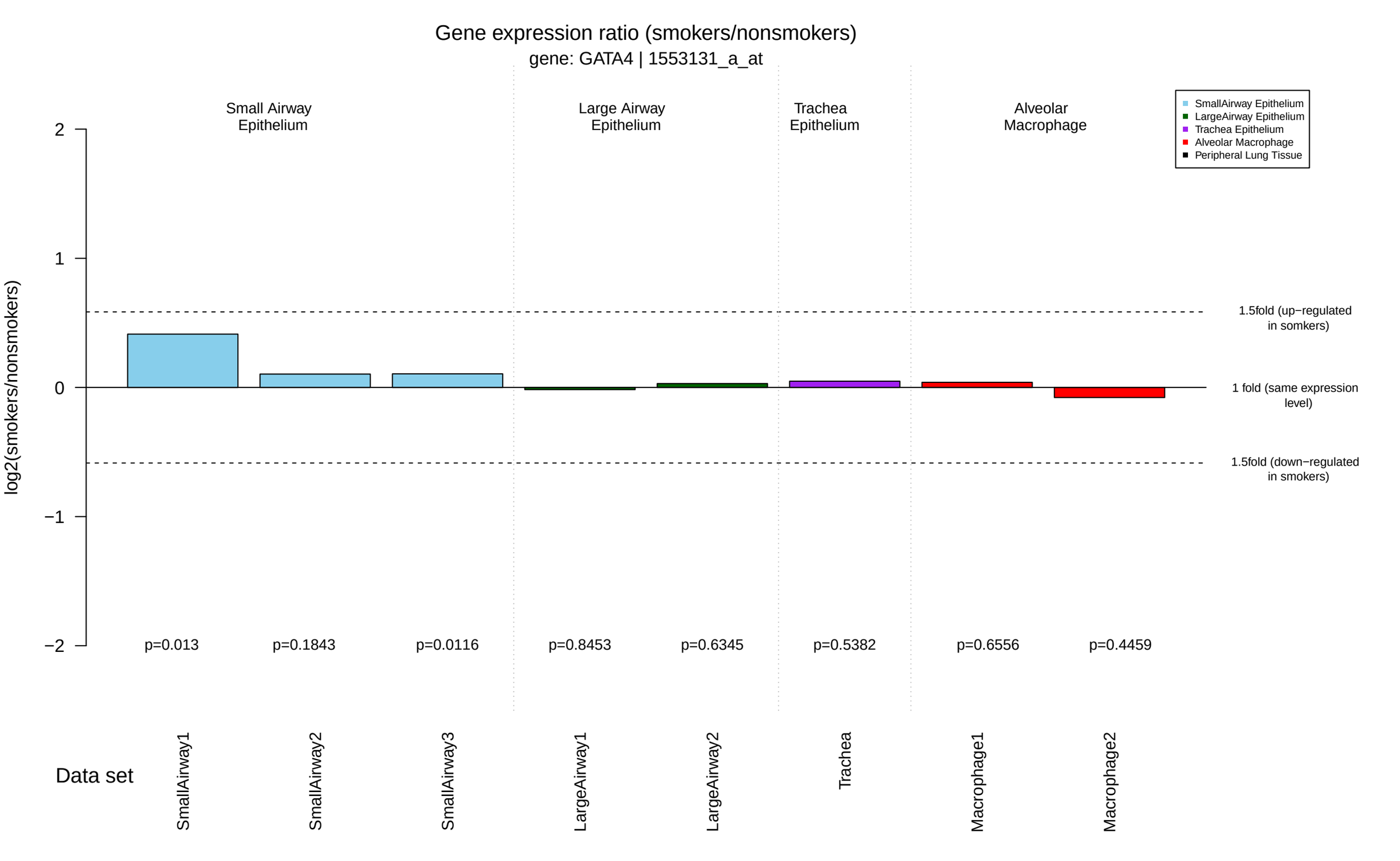
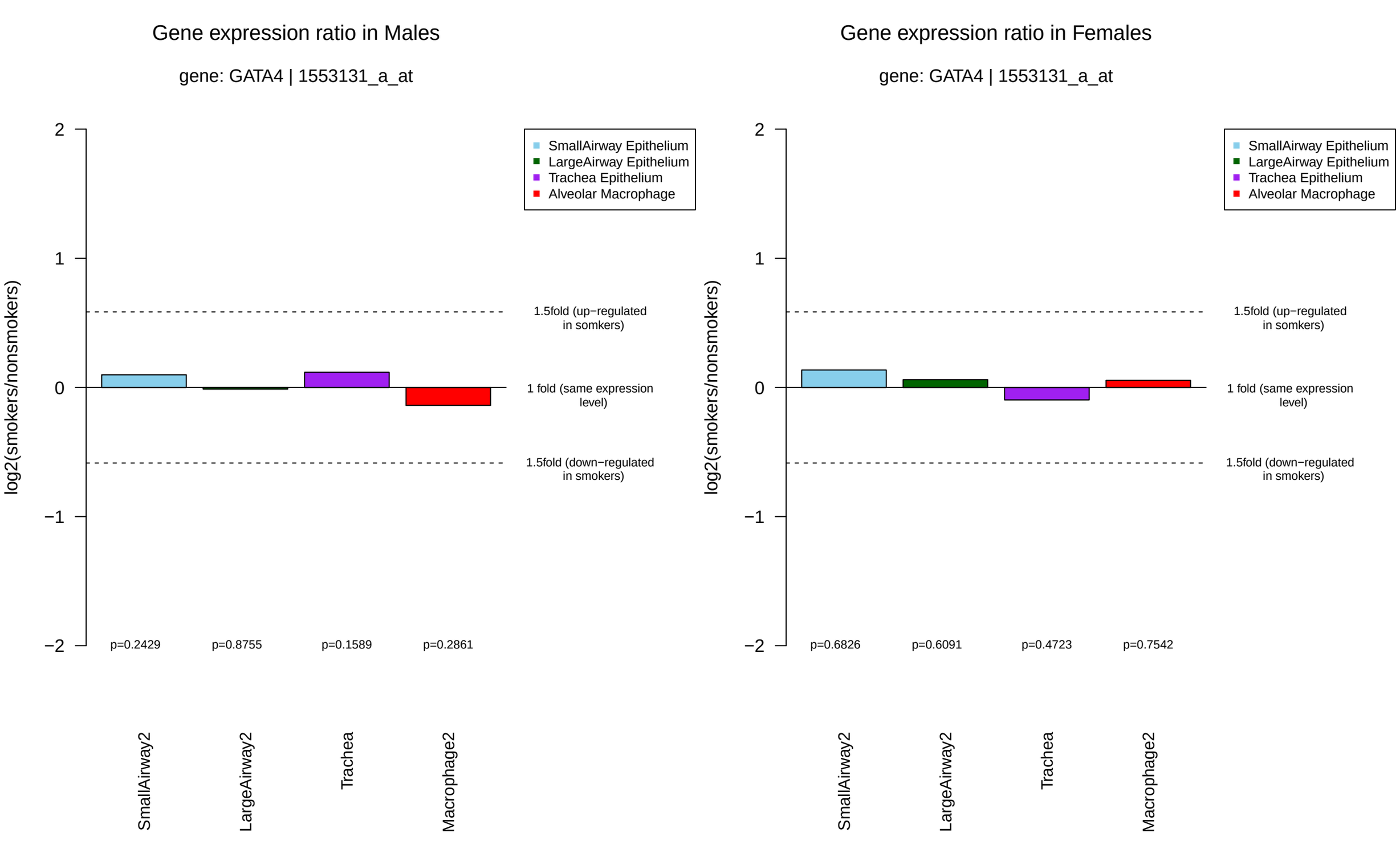
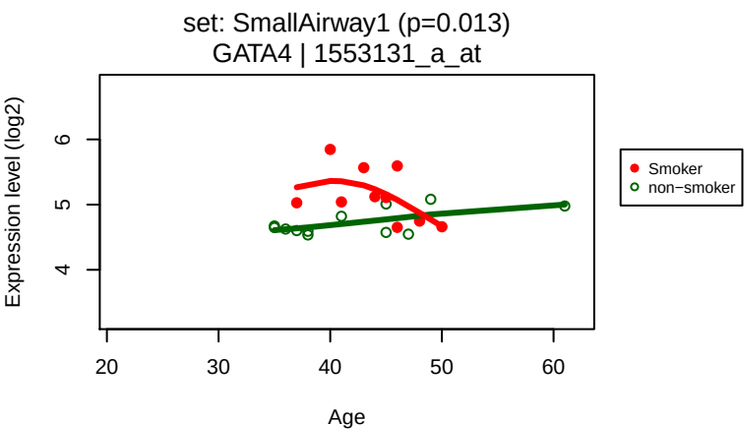
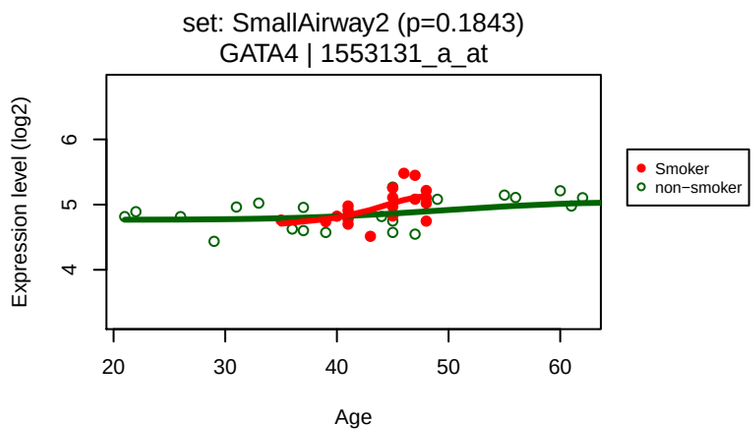
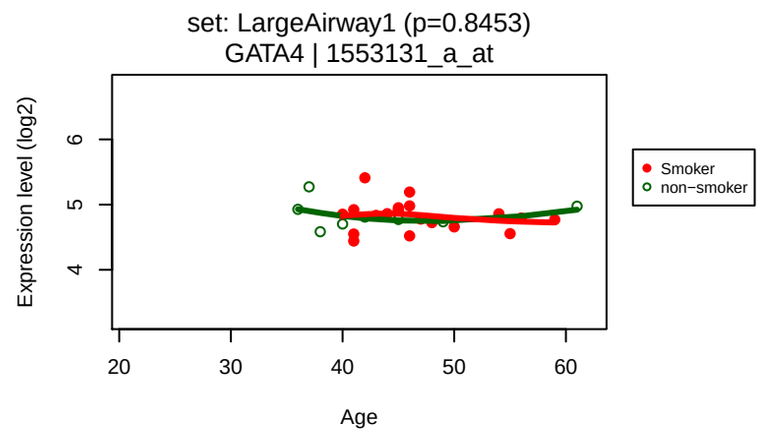

Gene expression of smokers and nonsmokers
gene: GATA4 | 1553131_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.013
p=0.1843
p=0.0116
p=0.8453
p=0.6345
p=0.5382
p=0.6556
p=0.4459
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

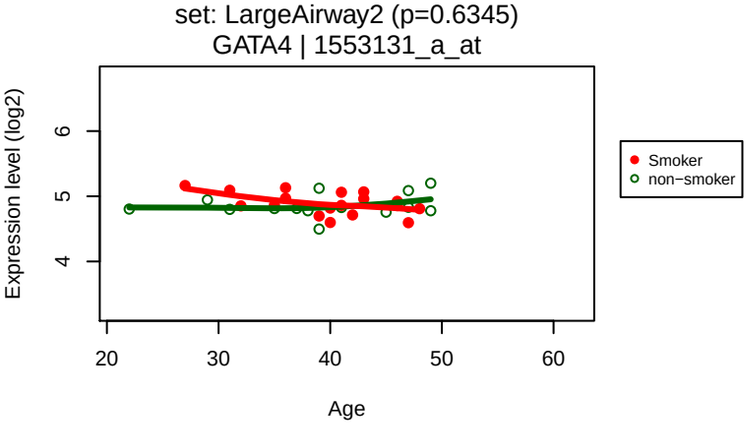

Gene expression ratio (smokers/nonsmokers)
gene: GATA4 | 1553131_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.013
p=0.1843
p=0.0116
p=0.8453
p=0.6345
p=0.5382
p=0.6556
p=0.4459
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

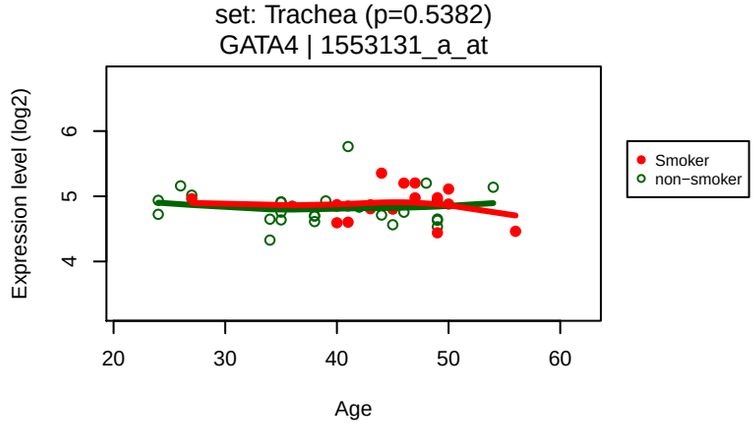

Gene expression ratio in Males
gene: GATA4 | 1553131_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2429
p=0.8755
p=0.1589
p=0.2861
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GATA4 | 1553131_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6826
p=0.6091
p=0.4723
p=0.7542
SmallAirway2
LargeAirway2
Trachea
Macrophage2

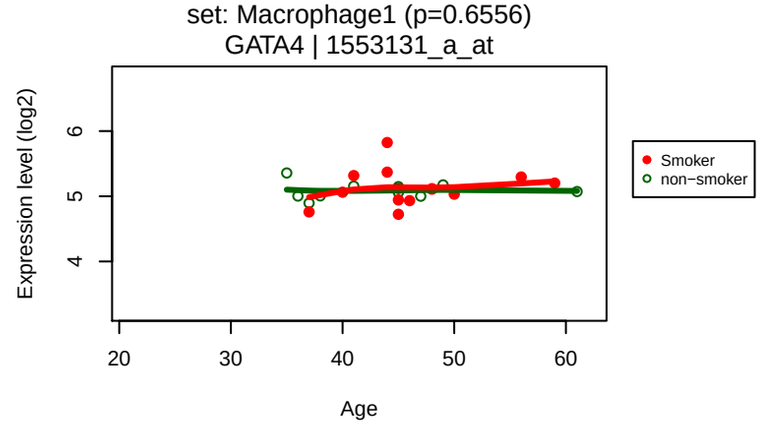

set: SmallAirway1 (p=0.013)
GATA4 | 1553131_a_at
Expression level (log2)
Age
Smoker
non−smoker

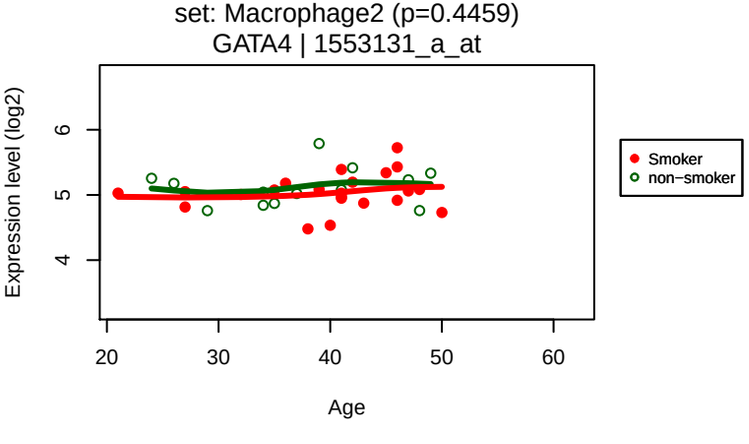

set: SmallAirway2 (p=0.1843)
GATA4 | 1553131_a_at
Expression level (log2)
Age
Smoker
non−smoker

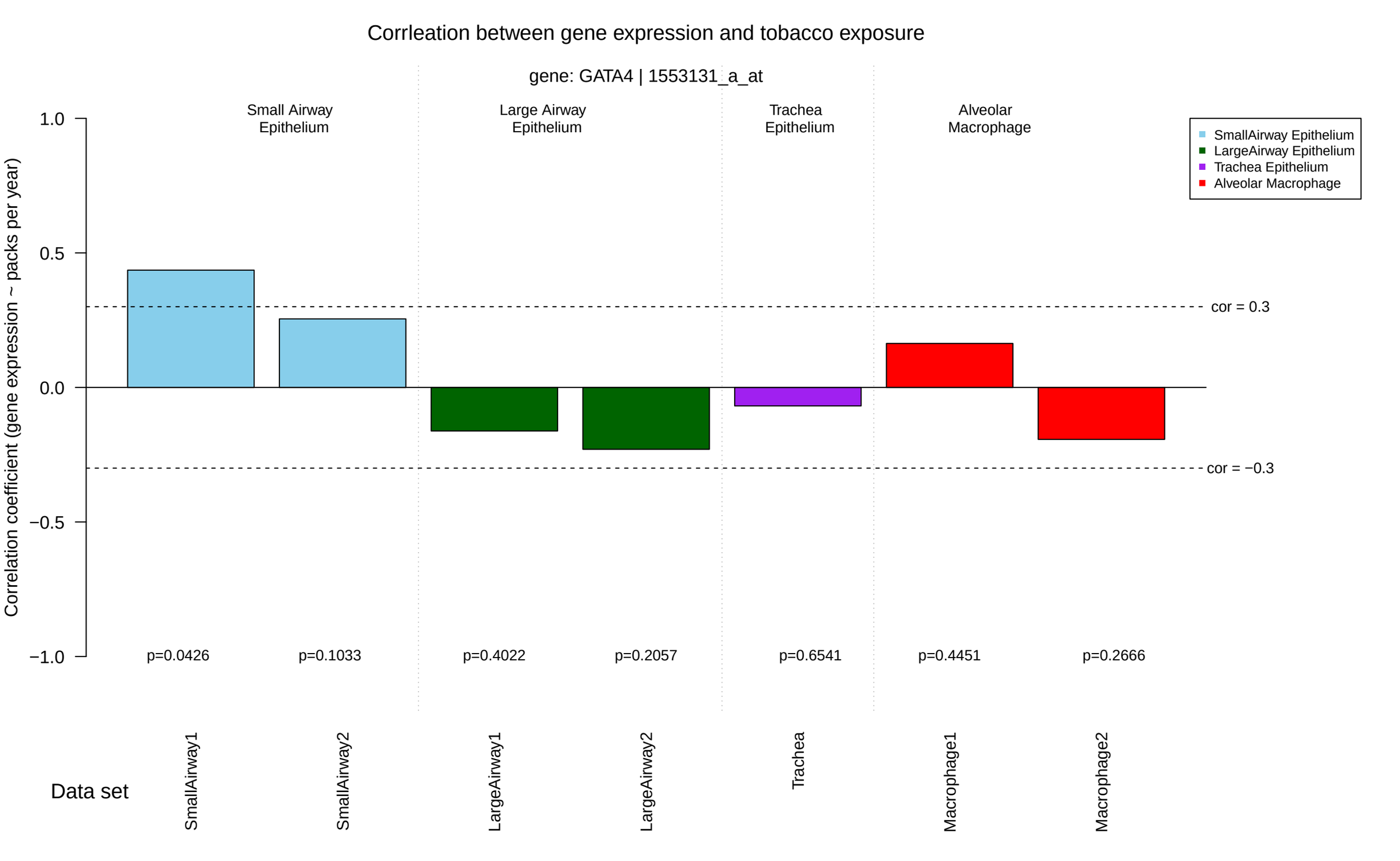

set: LargeAirway1 (p=0.8453)
GATA4 | 1553131_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6345)
GATA4 | 1553131_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5382)
GATA4 | 1553131_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6556)
GATA4 | 1553131_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4459)
GATA4 | 1553131_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GATA4 | 1553131_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0426
p=0.1033
p=0.4022
p=0.2057
p=0.6541
p=0.4451
p=0.2666
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2